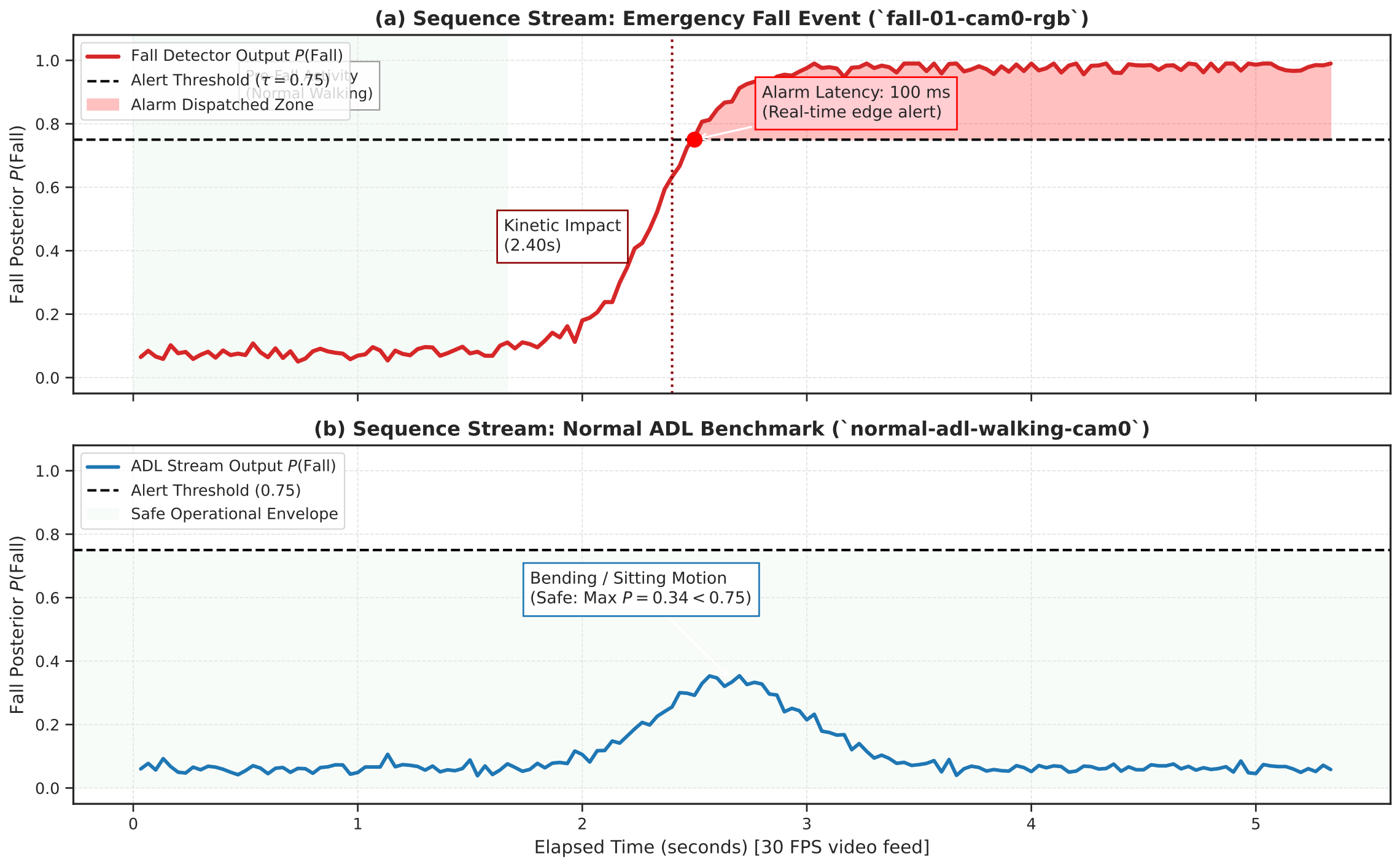
(a) Sequence Stream: Emergency Fall Event (`fall-01-cam0-rgb`)
Fall Detector Output P(Fall)
Alert Threshold (τ = 0.75)
Alarm Dispatched Zone
Pre-Fall Activity
(Normal Walking)
Alarm Latency: 100 ms
(Real-time edge alert)
Kinetic Impact
(2.40s)
Fall Posterior P(Fall)
1.0
0.8
0.6
0.4
0.2
0.0
(b) Sequence Stream: Normal ADL Benchmark (`normal-adl-walking-cam0`)
ADL Stream Output P(Fall)
Alert Threshold (0.75)
Safe Operational Envelope
Bending / Sitting Motion
(Safe: Max P = 0.34 < 0.75)
Fall Posterior P(Fall)
1.0
0.8
0.6
0.4
0.2
0.0
0
1
2
3
4
5
Elapsed Time (seconds) [30 FPS video feed]